

# JOSHUA NWACHINEMERE

AI Engineer | Python, Retrieval, Context Engineering, Multimodal AI & ML Evaluation

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## PROFILE

AI Engineer building Python systems around language, vision, and speech models. Work spans retrieval and context assembly, multimodal inputs, real-time transcription, model/API integration, FastAPI services, asynchronous workflows, and ML evaluation. Building VolyxAI with explicit data flow, validation, observability, and human control. Open to AI Engineer and ML Engineer opportunities.

## CORE SKILLS

- Languages: Python, SQL
- AI Engineering: retrieval, context engineering, multimodal inference, voice AI, structured outputs, provider routing, model evaluation
- Models & Providers: Azure AI Foundry, OpenAI, Anthropic, Gemini, DeepSeek, Deepgram
- ML & Data: scikit-learn, XGBoost, pandas, feature engineering, temporal evaluation
- Backend & Automation: FastAPI, asyncio, REST APIs, n8n, webhooks, SQLAlchemy
- Data & Infrastructure: PostgreSQL, SQLite, Redis, Docker, GitHub Actions, Linux
- Security: Suricata, Snort, VirusTotal API, OSINT, DNS analysis, secure credential management

## INDEPENDENT PRODUCT & ENGINEERING WORK

### AI Engineer · VolyxAI product work | Nov 2025 - Present

*Independent product effort*

- Developing applied AI systems that combine model and provider integration, retrieval and context assembly, voice and multimodal workflows, and structured outputs.
- Building Python services and event-driven n8n workflows around external APIs, webhooks, validation, retries, logging, and human approval.
- Exploring provider routing, fallback, cost, latency, and privacy trade-offs across Azure AI Foundry, OpenAI-compatible APIs, and transcription providers.

**Key tools:** Python, FastAPI, n8n, REST APIs, webhooks, Azure AI Foundry, OpenAI-compatible APIs, PostgreSQL, Redis

### AI, Backend & Automation Projects | Jan 2021 - Present

*Independent engineering work*

- Built Python automation and backend utilities for data processing, Telegram bots, API integrations, and operational tooling.
- Created asynchronous VirusTotal scanning and multi-chain wallet analytics with validation, caching, pagination, and rate-limit handling.
- Developed and evaluated an experimental football forecasting pipeline with temporal features, calibrated models, FastAPI, and rolling-origin testing.

**Key tools:** Python, Async I/O, Telegram Bot API, aiohttp, SQL, Docker, REST APIs

## TECHNICAL TRAINING

### Threat Detection & Response Capstone | Capstone IT Staffing (Remote) | Jul 2025 - Aug 2025

- Deployed Suricata on Ubuntu, wrote custom Snort rules, and automated alert enrichment with Python and public OSINT sources.
- Built a DNS intelligence CLI with asynchronous resolution, WHOIS lookups, trace output, and JSON export.

## OPEN-SOURCE CONTRIBUTIONS

### OpenAI Agents Python SDK | Merged PR #3991

- Made transient pre-response WebSocket server errors follow the SDK's retry policy while preserving non-transient handling and replay-safety checks.
- PR: <https://github.com/openai/openai-agents-python/pull/3991>

### Pydantic AI Harness | Merged PR #503

- Surfaced unexpected Code Mode execution failures to the model as retries after resetting invalid session state, enabling recovery from lost imports and variables.
- PR: <https://github.com/pydantic/pydantic-ai-harness/pull/503>

### Mellea | Merged PR #1471

- Added deterministic mocked-backend tests for async span timing, context propagation, token usage, span lifetime, and consecutive generations.
- PR: <https://github.com/generative-computing/mellea/pull/1471>

### FastStream | Merged PR #2961

- Restored FastAPI 0.140 compatibility for slotted Dependant objects while retaining native dependency fields and FastStream schema metadata.
- PR: <https://github.com/ag2ai/faststream/pull/2961>

### Additional verified upstream merges | 4 of 8 merged PRs

- Apache Arrow Rust: <https://github.com/apache/arrow-rs/pull/10486> | Altair: <https://github.com/vega/altair/pull/4089> | FastStream FastAPI: [https://github.com/faststream-community/faststream\\_fastapi/pull/2](https://github.com/faststream-community/faststream_fastapi/pull/2) | Calkit: <https://github.com/calkit/calkit/pull/1028>

## SELECTED PROJECTS

### Volyx Lens | Privacy-First Context-Aware AI Assistant

*JavaScript, Electron, Swift, Azure AI Foundry, Multimodal APIs*

- Built and hardened a privacy-oriented macOS assistant combining selected screen context, microphone input, meeting audio, local OCR, relevance-ranked retrieval, and multiple AI providers.
- Implemented bounded context memory, transcription and provider routing, explicit data-sharing controls, and cost-aware fallbacks.
- Hardened Electron boundaries with sandboxing, context isolation, restricted IPC, automated tests, secret scanning, and release checks.
- Source: <https://github.com/dk3yyyy/volyx-lens>

### Football Predictor | Experimental ML Pipeline

*Python, XGBoost, scikit-learn, FastAPI, Streamlit, SQLAlchemy*

- Built data scrapers, temporal feature engineering, XGBoost outcome and Poisson goal models, a FastAPI service, and a Streamlit dashboard.
- Evaluated with rolling-origin testing across 1,140 Premier League matches: 53.77% outcome accuracy versus a 56.70% bookmaker-implied benchmark, showing signal but no practical betting advantage.
- Source: [https://github.com/dk3yyyy/football\\_predictor](https://github.com/dk3yyyy/football_predictor)

### Noughtline | Real-Time Multiplayer System

*React, Express, Socket.IO, SQLite, Paystack*

- Built signed guest sessions, server-authoritative moves, reconnect and forfeit handling, transactional match settlement, and an append-only currency ledger.
- Added idempotent Paystack crediting plus automated API, economy, room-state, and two-client Socket.IO tests.
- Live: <https://noughtline.onrender.com> | Source: <https://github.com/dk3yyyy/Noughtline>

## CERTIFICATIONS

- Google AI Professional Certificate | Coursera
- Google Cybersecurity Professional Certificate | Coursera

## EDUCATION

Bachelor of Technology (BTech), Mathematics | 2016 - 2021

Federal University of Technology, Owerri (FUTO)